Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

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Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw Generating Circle & Directing Line (Base)

C_0 $\longrightarrow$ Initial Center Point
AA' $\longrightarrow$ Directing Line
Axis $\longrightarrow$ Center Line

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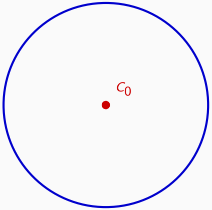

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